

ERGA Assembly Report

v24.10.15

Tags: ERGA-BGE

TxID	166086
ToLID	rHemHip1
Species	Hemorrhois hippocrepis
Class	Lepidosauria
Order	Squamata

Genome Traits	Expected	Observed
Haploid size (bp)	1,677,182,640	1,851,468,933
Haploid Number	18 (source: ancestor)	19
Ploidy	3 (source: ancestor)	2
Sample Sex	ZW	ZW

EBP metrics summary and curation notes

Obtained EBP quality metric for hap1: 8.8.Q66

Obtained EBP quality metric for hap2: 7.8.Q66

The following metrics were automatically flagged as below EBP recommended standards or different from expected:

- . Observed Haploid Number is different from Expected
- . Observed Ploidy is different from Expected
- . Kmer completeness value is less than 90 for hap2
- . Assembly length loss > 3% for hap2

Curator notes

- . Interventions/Gb: 20
- . Contamination notes: "No contaminants were detected"
- . Other observations: "The reads were re-assembled with hifiasm (ont) with Hi-C phasing (initially with CLAWS 3.0 and finishing with remapping and evaluation steps with CLAWS 3.3). In the initial round of curation, we made 2 cuts in contigs, 10 breaks at gaps and 8 joins. The -better- SUPERS (higher BUSCO, fewer gaps, longer length) as well as the Z and W chrs were assigned to hap1 a posteriori."

Quality metrics table

Metrics	Pre-curation hap1	Pre-curation hap2	Curated hap1	Curated hap2
Total bp	1,689,233,707	1,696,562,687	1,851,468,933	1,531,310,509
GC %	41.19	41.06	41.18	41.06
Gaps/Gbp	13.02	8.84	10.26	11.1
Total gap bp	4,400	3,000	3,800	3,400
Scaffolds	57	29	55	32
Scaffold N50	223,150,264	220,572,697	154,801,275	220,572,697
Scaffold L50	3	3	4	3
Scaffold L90	12	9	11	10
Contigs	79	44	74	49
Contig N50	94,437,553	132,354,085	114,638,990	93,867,649
Contig L50	6	5	6	5
Contig L90	20	15	21	16
QV	66.1566	67.263	66.566	66.8302
Kmer compl.	88.7141	93.5576	95.5531	85.7345
BUSCO sing.	91.2%	96.2%	97.4%	92.4%
BUSCO dupl.	0.8%	0.8%	1.3%	0.8%
BUSCO frag.	0.7%	0.8%	0.1%	0.1%
BUSCO miss.	7.3%	2.2%	1.1%	6.6%

Warning! BUSCO versions or lineage datasets are not the same across results:

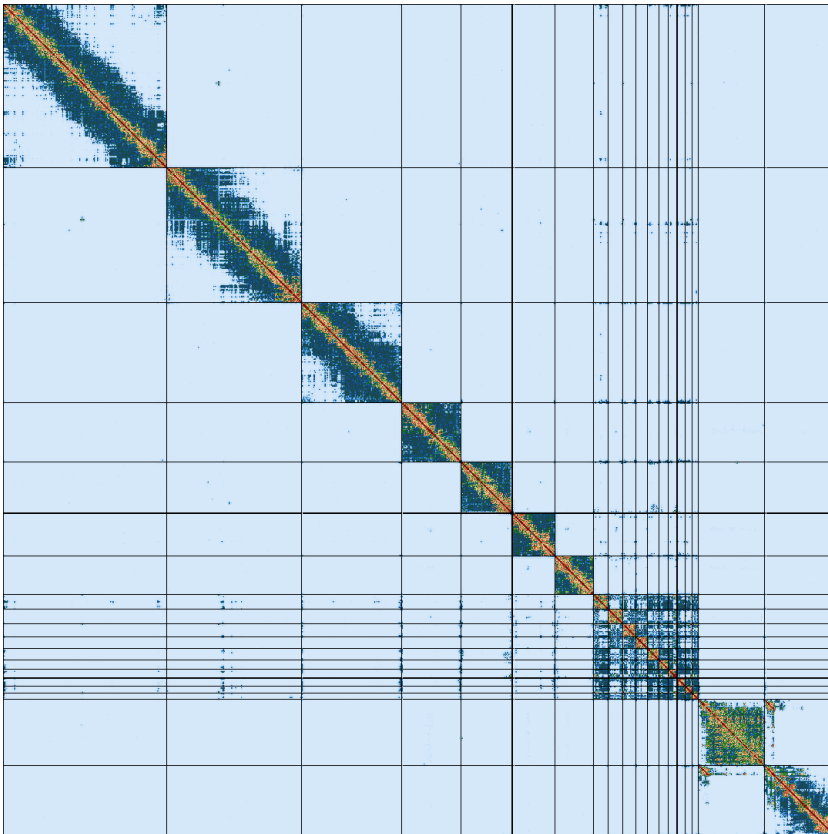
BUSCO: 5.5.0 (euk_genome_met, metaeuk) / Lineage: vertebrata_odb10 (genomes:67, BUSCOs:3354)

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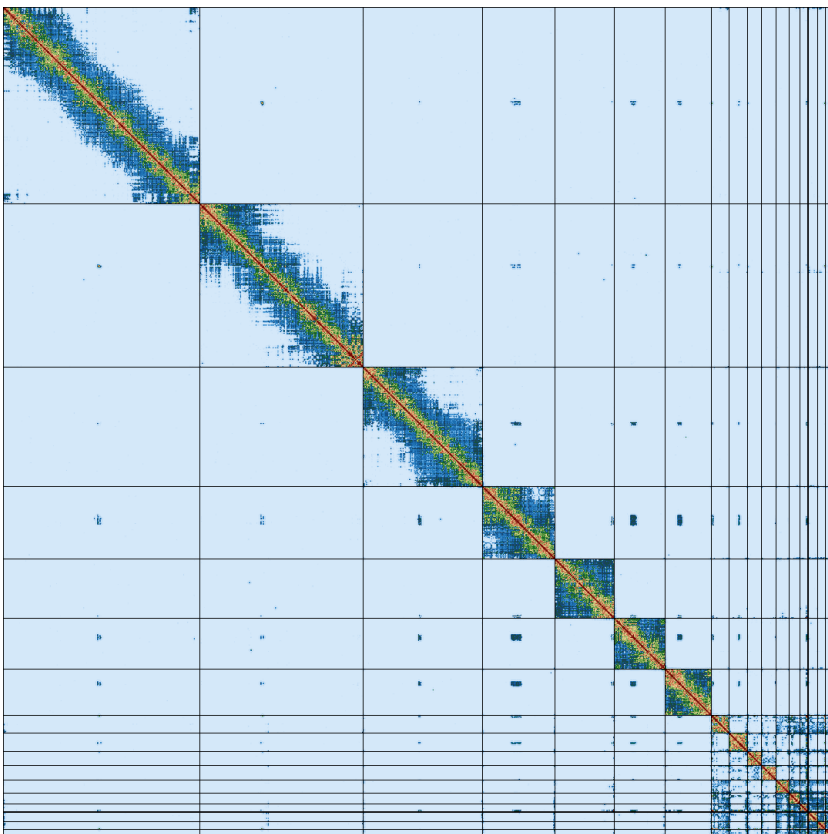
BUSCO: 6.0.0 (euk_genome_min, miniprot) / Lineage: vertebrata_odb10 (genomes:67, BUSCOs:3354)

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HiC contact map of curated assembly

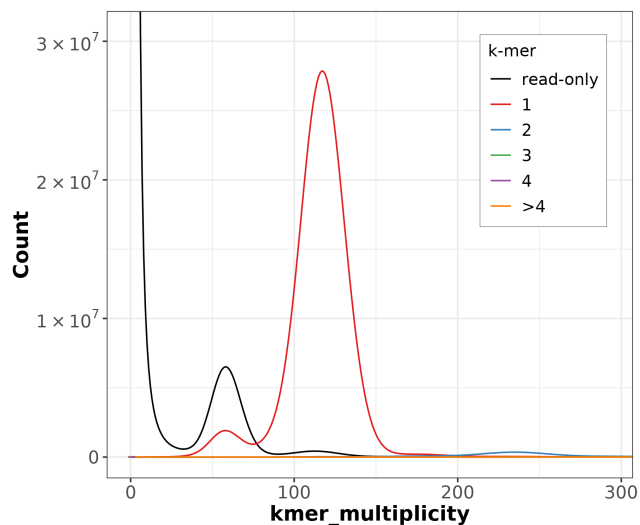


hap1 [\[LINK\]](#)

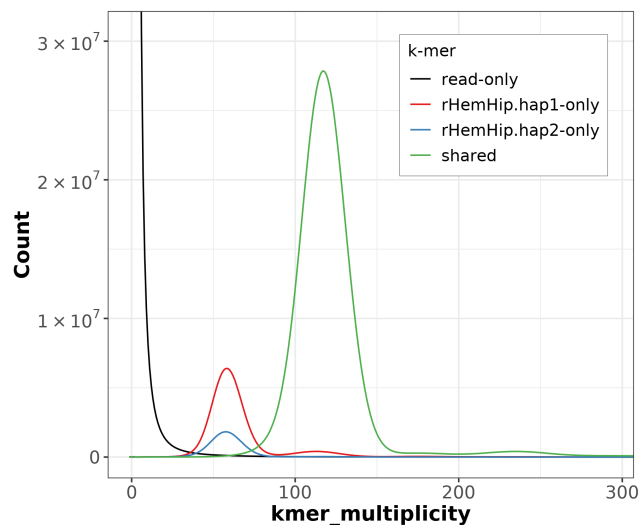


hap2 [\[LINK\]](#)

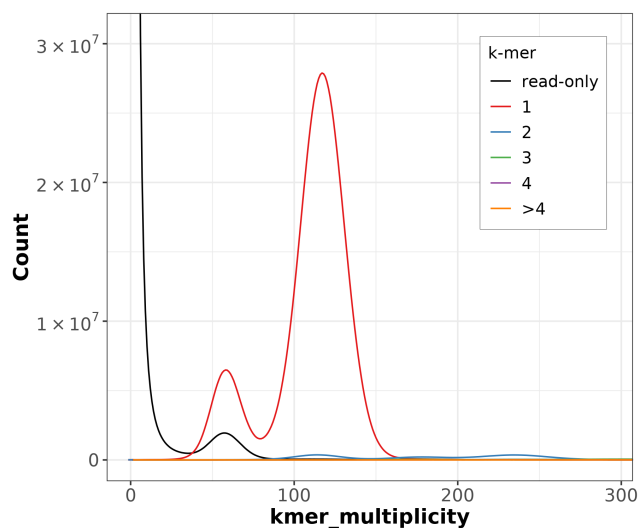
K-mer spectra of curated assembly



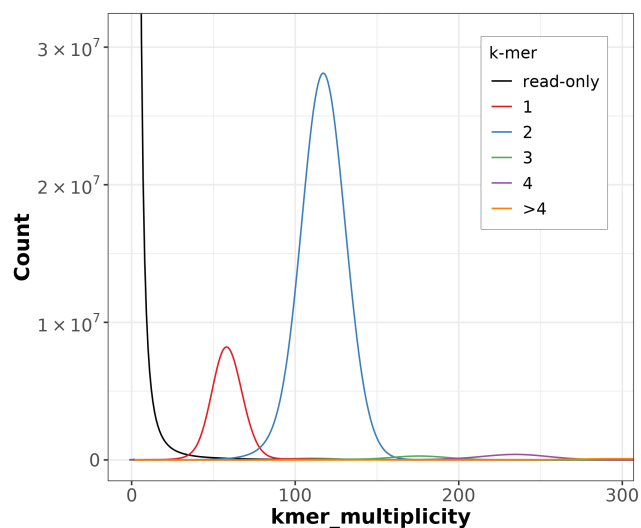
Distribution of k-mer counts per copy numbers found in **hap2** (hapl.)



Distribution of k-mer counts coloured by their presence in reads/assemblies

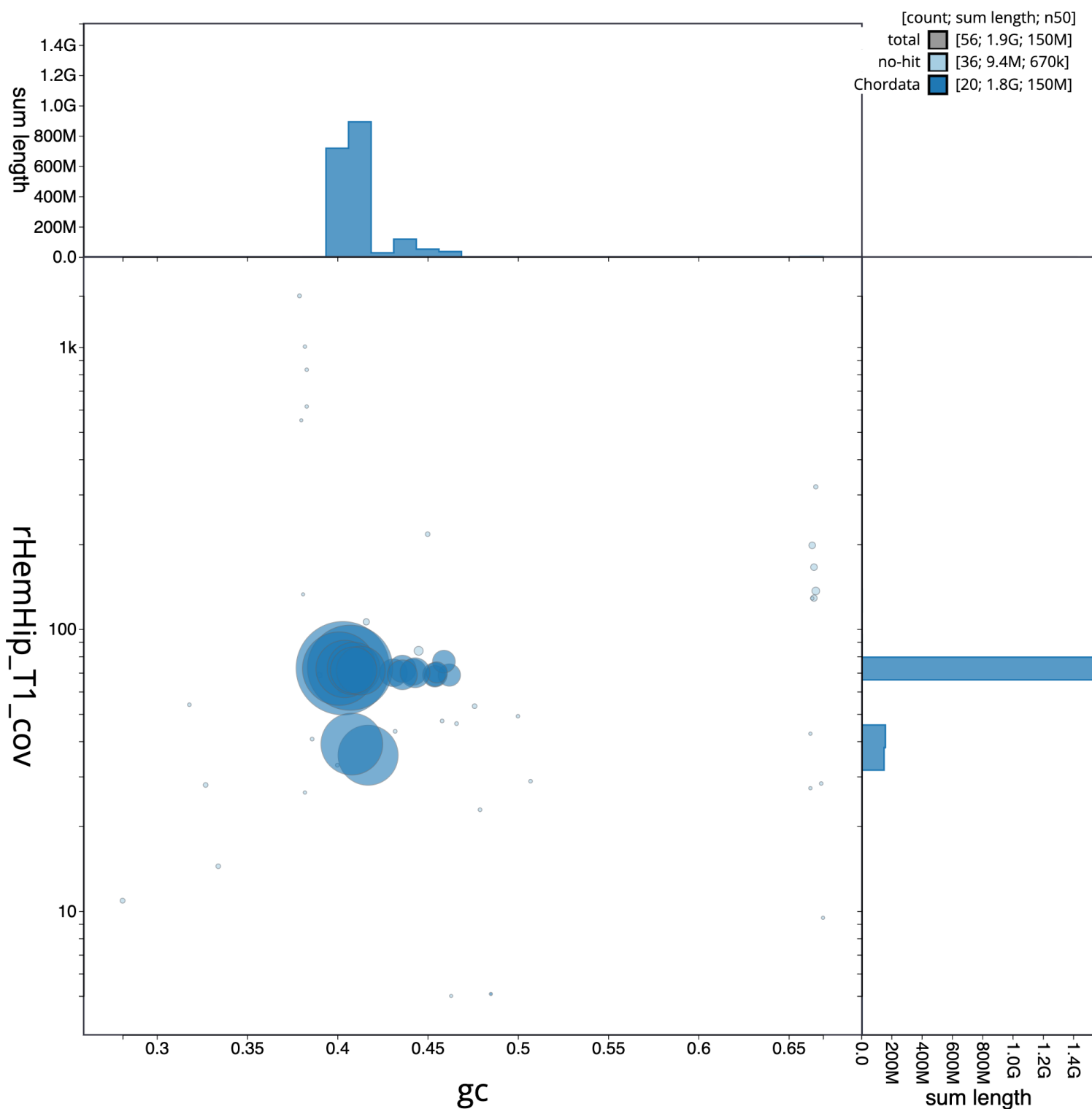


Distribution of k-mer counts per copy numbers found in **hap1** (hapl.)

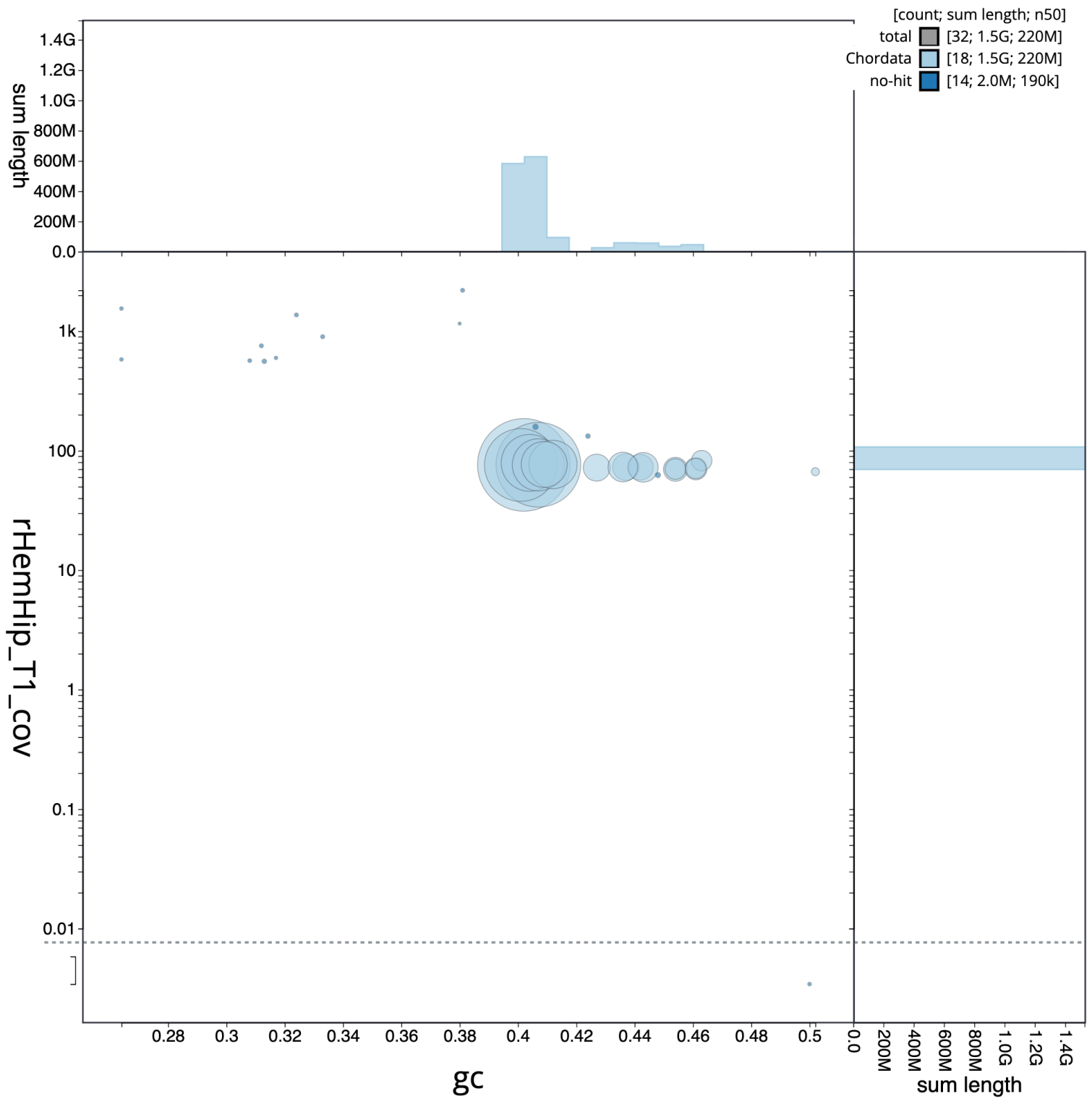


Distribution of k-mer counts per copy numbers found in **asm** (dipl.)

Post-curation contamination screening



hap1. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.



hap2. Bubble plot circles are scaled by sequence length, positioned by coverage and GC proportion, and coloured by taxonomy. Histograms show total assembly length distribution on each axis.

Data profile

Data	ONT	Illumina	OmniC
Coverage	84x	60x	53x

Assembly pipeline

- **CLAWS**
 - |_ *ver*: v3.3.0
 - |_ *key param*: NA
- **Filtlong**
 - |_ *ver*: v0.2.1
 - |_ *key param*: NA
- **Hifiasm**
 - |_ *ver*: 0.24.0
 - |_ *key param*: --ont
- **Yahs**
 - |_ *ver*: 1.2a
 - |_ *key param*: NA
- **Chromap**
 - |_ *ver*: 0.3.2-r518
 - |_ *key param*: NA

Curation pipeline

- **Blobtoolkit Nextflow pipeline**
 - |_ *ver*: v0.6
 - |_ *key param*: NA
- **GRIT_Rapid**
 - |_ *ver*: 2.0
 - |_ *key param*: NA
- **PretextViewAI**
 - |_ *ver*: 1.0.7
 - |_ *key param*: NA

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